

INDIA HOUSEHOLD WELLBEING DASHBOARD

"This project is an Executive Power BI Dashboard built using India's MOSPI Household Consumption Expenditure Survey 2024.

The objective was to convert more than 224,000 household records into an interactive decision-support dashboard for policymakers.

I followed a complete BI workflow including Python data preprocessing, star schema modeling, DAX measures, Power Query transformation, and finally developed a five-page executive dashboard."

Data Model (Explain Before Dashboard)

"I first created a Star Schema because it is the best practice for Power BI."

Fact Table

- Fact_Household

Dimension Tables

- Dim_State
- Dim_District
- Dim_Sector
- Dim_Wellbeing
- Dim_Vulnerability

Supporting Tables

- State KPI
- District KPI
- National KPI

"This structure reduces redundancy and improves report performance."

Page 1 — Executive Overview

"The first page gives management a quick overview of India's household wellbeing."

Explain each section:

KPI Cards

"These cards summarize the national condition."

- Total Households

- Population Weight
 - Average Wellbeing
 - Food Index
 - Vulnerability
 - Food Inflation
-

Wellbeing Distribution

"This donut chart shows how households are distributed across wellbeing classes."

Meaning

- balanced distribution
 - no extreme concentration
-

Vulnerability Distribution

"This chart immediately shows that most households belong to Low or Moderate vulnerability."

India Map

"The filled map highlights geographical differences in wellbeing."

Decision

Which states require more attention.

Top Performing States

"This ranking identifies the highest performing states."

Population Treemap

"This visual helps compare population concentration across states."

Bottom Observations

"Instead of making users interpret every chart, I summarized the key insights for executives."

Page 2 — State Performance Analytics

"This page allows comparison of states using multiple socioeconomic indicators."

KPI Cards

Average

- Wellbeing
- Economic Index
- Food Index
- Asset Index
- Vulnerability

Performance Table

"This table allows comparison of all states using every important indicator."

Top States

"Shows the best-performing states."

Scatter Plot

Economic Index vs Food Index

Explain

"This shows the relationship between economic wellbeing and food wellbeing."

Higher-right

Good

Lower-left

Poor

Decomposition Tree

Explain

"This is one of Power BI's AI visuals."

"I can drill from

State – Sector – Wellbeing -- Vulnerability

It automatically explains

Why a state performs well.

Page 3 — District Intelligence Explorer

"The previous page was state-level.

This page goes deeper to district-level."

Explain

District Count

Average Wellbeing

Population

Top Districts

Bottom Districts

Scatter Plot

District Economic vs Food

District Table

"Allows district-wise comparison."

Treemap

Population Distribution

Decision

District level planning.

Page 4 — Household Vulnerability Assessment

"This page focuses on identifying risk."

Explain KPIs

High Risk Households

Average Vulnerability

Low Wellbeing Households

Luxury Households

Housing Burden

Scatter

Wellbeing vs Vulnerability

"Households with low wellbeing and high vulnerability need immediate support."

Urban vs Rural

"Shows sector differences."

Top Vulnerable States

Household Table

Used for detailed investigation.

Decision

Government welfare targeting.

Page 5 — Policy Insights & Decision Support

"This is the most important page because it converts analysis into recommendations."

Explain

Policy Priority

High Risk Households

Best State

Worst State

Top Policy States

States needing intervention.

Top Policy Districts

District-level priorities.

Policy Risk Matrix

Shows

Wellbeing vs Vulnerability

Key Driver Ranking

Very important.

"I calculated Pearson correlation in Python to identify the strongest drivers of wellbeing."

Results

Economic Index ≈ 0.99

Asset Index ≈ 0.83

Food Index ≈ 0.78

Welfare Index ≈ 0.78

Housing Burden ≈ 0.35

Food Inflation ≈ 0.02

General Inflation ≈ -0.05

Explain

"This tells policymakers that improving economic conditions and household assets has a much larger impact than short-term inflation changes."

Recommendation Table

Shows

State

Wellbeing

Vulnerability

Housing Burden

Food Index

Decision

Bottom Key Observations

"These summarize the entire dashboard into actionable recommendations."

Navigation

Point this out proudly.

"I also designed the dashboard to improve user experience."

Features

- ✓ Left Navigation Bar
 - ✓ Filter Panel
 - ✓ Hide/Show Filter
 - ✓ Clear Filters Button
 - ✓ Cross-page filtering
 - ✓ Consistent theme
-

End the Presentation

"Overall, this dashboard transforms raw household survey data into a policy decision-support system.

Instead of just displaying statistics, it identifies where intervention is needed, what factors influence wellbeing the most, and helps decision-makers prioritize states and districts using data."

